
Spring-boot

By adsoft

Spring-boot

Install mvn

<https://maven.apache.org/>

1. Update package list:

```
sudo apt update
```

2. Install Maven:

```
sudo apt install maven -y
```

3. Verify installation:

```
mvn -version
```

```
adsoft@fnt-ori213:~$ mvn --version
```

```
Apache Maven 3.9.9
```

```
Maven home: /usr/share/maven
```

```
Java version: 25.0.2, vendor: Ubuntu, runtime: /usr/lib/jvm/java-25-openjdk-amd64
```

```
Default locale: en_US, platform encoding: UTF-8
```

```
OS name: "linux", version: "6.17.0-14-generic", arch: "amd64", family: "unix"
```

```
adsoft@fnt-ori213:~$
```

Spring-boot

Install sdkman

<https://sdkman.io/>

```
adsoft@fnt-ori213:~$ sdk version
```

SDKMAN!

script: 5.20.0

native: 0.7.21 (linux x86_64)

```
adsoft@fnt-ori213:~$
```

1. Update your system to ensure all packages are up-to-date:

```
sudo apt update && sudo apt upgrade -y
```

2. Install required dependencies (curl, zip, unzip):

```
sudo apt install -y curl zip unzip
```

3. Download and run the SDKMAN installation script:

```
curl -s "https://get.sdkman.io" | bash
```

4. Initialize SDKMAN in your current shell session:

```
source "$HOME/.sdkman/bin/sdkman-init.sh"
```

5. Verify the installation by checking the SDKMAN version:

```
sdk version
```

Install spring-boot

Install Spring Boot CLI

Use SDKMAN! to install the Spring Boot CLI:

```
sdk install springboot
```

Verify the Installation

Confirm the installation was successful by checking the version:

```
spring --version
```

```
adsoft@fnt-ori213:~$ spring --version
Spring CLI v4.0.3
adsoft@fnt-ori213:~$
```

Create spring-boot project <https://start.spring.io/>



start.spring.io

Install



Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ Java ☐ Kotlin ☐ Groovy

☒ Maven

Language

Spring Boot

☐ 4.1.0 (SNAPSHOT) ☐ 4.1.0 (M2) ☐ 4.0.4 (SNAPSHOT) ☒ 4.0.3

☐ 3.5.12 (SNAPSHOT) ☐ 3.5.11

Project Metadata

Group Artifact

Name

Description

Package name

Packaging ☒ Jar ☐ War

Configuration ☒ Properties ☐ YAML

Java ☐ 25 ☒ 21 ☐ 17

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Web WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

PostgreSQL Driver SQL

A JDBC and R2DBC driver that allows Java programs to connect to a PostgreSQL database using standard, database independent Java code.

Spring Data JPA SQL

Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.

GENERATE CTRL + ⌘

EXPLORE CTRL + SPACE

...

Unzip

demopg.zip

project

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton$ ls *.zip
demopg.zip
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton$ unzip demopg.zip
Archive:  demopg.zip
  creating: demopg/
  inflating: demopg/mvnw
  inflating: demopg/.gitattributes
  inflating: demopg/mvnw.cmd
  inflating: demopg/pom.xml
   creating: demopg/src/
   creating: demopg/src/main/
   creating: demopg/src/main/java/
   creating: demopg/src/main/java/com/
   creating: demopg/src/main/java/com/postgres/
   creating: demopg/src/main/java/com/postgres/demopg/
  inflating: demopg/src/main/java/com/postgres/demopg/DemopgApplication.java
   creating: demopg/src/main/resources/
  inflating: demopg/src/main/resources/application.properties
   creating: demopg/src/main/resources/static/
   creating: demopg/src/main/resources/templates/
   creating: demopg/src/test/
   creating: demopg/src/test/java/
   creating: demopg/src/test/java/com/
   creating: demopg/src/test/java/com/postgres/
   creating: demopg/src/test/java/com/postgres/demopg/
  inflating: demopg/src/test/java/com/postgres/demopg/DemopgApplicationTests.java
   creating: demopg/.mvn/
   creating: demopg/.mvn/wrapper/
  inflating: demopg/.mvn/wrapper/maven-wrapper.properties
  inflating: demopg/.gitignore
  inflating: demopg/HELP.md
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton$
```

Set properties

```
$ cd demopg
```

```
$ vi src/main/resources/application.properties
```

```
spring.application.name=demopg
spring.application.name=SpringBootSecurityPostgresqlApplication
# spring.datasource.url=jdbc:mysql://localhost:3306/testdb_spring?useSSL=false
spring.datasource.url=jdbc:postgresql://localhost:5432/user_api?useSSL=false
spring.datasource.username=postgres
spring.datasource.password=adsoft
spring.jpa.database=POSTGRESQL
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.PostgreSQLDialect
spring.jpa.hibernate.ddl-auto=update
```

mvn install

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ mvn install
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.google.common.util.concurrent.AbstractFuture$UnsafeAtomicHelper (file:/usr/share/maven/lib/guava.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.common.util.concurrent.AbstractFuture$UnsafeAtomicHelper
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.postgres:demopg >-----
[INFO] Building demopg 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
2/maven-resolver-api-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom (2.2 kB at 26 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar (196 kB at 646 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (157 kB at 704 kB/s)
[INFO] Installing /home/adsoft/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/pom.xml to /home/adsoft/.m2/repository/com/postgres/demopg/0.0.1-SNAPSHOT/demopg-0.0.1-SNAPSHOT.pom
[INFO] Installing /home/adsoft/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/target/demopg-0.0.1-SNAPSHOT.jar to /home/adsoft/.m2/repository/com/postgres/demopg/0.0.1-SNAPSHOT/demopg-0.0.1-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 29.122 s
[INFO] Finished at: 2026-03-05T18:30:58-06:00
[INFO] -----
```

mvn spring-boot:run

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ mvn spring-boot:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.google.common.util.concurrent.AbstractFuture$UnsafeAtomicHelper (
file:/usr/share/maven/lib/guava.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.common.util.concurrent.AbstractFuture$UnsafeAtomicHel
per
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.postgres:demopg >-----
[INFO] Building demopg 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:4.0.3:run (default-cli) > test-compile @ demopg >>>

ons.pooling      : HHH10001005: Database info:
    Database JDBC URL [jdbc:postgresql://localhost:5432/user_api?useSSL=false]
    Database driver: PostgreSQL JDBC Driver
    Database dialect: PostgreSQLDialect
    Database version: 17.9
    Default catalog/schema: user_api/public
    Autocommit mode: undefined/unknown
    Isolation level: READ_COMMITTED [default READ_COMMITTED]
    JDBC fetch size: none
    Pool: DataSourceConnectionProvider
    Minimum pool size: undefined/unknown
    Maximum pool size: undefined/unknown
2026-03-05T18:36:40.805-06:00 INFO 20802 --- [SpringBootSecurityPostgresqlApplication] [      main] org.hibernate.orm.core
    : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-03-05T18:36:40.809-06:00 INFO 20802 --- [SpringBootSecurityPostgresqlApplication] [      main] j.LocalContainerEntityMana
gerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-03-05T18:36:40.830-06:00 WARN 20802 --- [SpringBootSecurityPostgresqlApplication] [      main] JpaBaseConfiguration$JpaWe
bConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering.
Explicitly configure spring.jpa.open-in-view to disable this warning
2026-03-05T18:36:40.965-06:00 INFO 20802 --- [SpringBootSecurityPostgresqlApplication] [      main] o.s.boot.tomcat.TomcatWebS
erver : Tomcat started on port 8080 (http) with context path '/'
2026-03-05T18:36:40.967-06:00 INFO 20802 --- [SpringBootSecurityPostgresqlApplication] [      main] com.postgres.demopg.Demopg
Application : Started DemopgApplication in 1.502 seconds (process running for 1.662)
```

Test : localhost:8080



localhost:8080



Whitelabel Error Page

This application has no explicit mapping for /error, so you are seeing this as a fallback.

Thu Mar 05 18:39:41 CST 2026

There was an unexpected error (type=Not Found, status=404).

Update pom.xml

```
<dependency>  
  <groupId>jakarta.validation</groupId>  
  <artifactId>jakarta.validation-api</artifactId>  
  <version>3.0.2</version>  
</dependency>
```

```
<dependency>  
  <groupId>org.springframework.boot</groupId>  
  <artifactId>spring-boot-starter-webmvc</artifactId>  
</dependency>
```

\$ mvn install

Create folders:

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/src/main/java/com/postgres/demopg$ mkdir models
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/src/main/java/com/postgres/demopg$ mkdir controllers
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/src/main/java/com/postgres/demopg$ mkdir repository
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/src/main/java/com/postgres/demopg$ ls -la
total 24
drwxr-xr-x 5 adsoft adsoft 4096 Mar  6 09:18 .
drwxr-xr-x 3 adsoft adsoft 4096 Mar  5 18:15 ..
-rw-r--r-- 1 adsoft adsoft  312 Mar  5 18:15 DemopgApplication.java
drwxrwxr-x 2 adsoft adsoft 4096 Mar  6 09:18 controllers
drwxrwxr-x 2 adsoft adsoft 4096 Mar  6 09:18 models
drwxrwxr-x 2 adsoft adsoft 4096 Mar  6 09:18 repository
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg/src/main/java/com/postgres/demopg$
```

Create Tweet Model:

\$ vi models/Tweet.java

```
package com.postgres.demopg.models;
import java.util.HashSet;
import java.util.Set;
import jakarta.persistence.*;
import jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.Size;

@Entity
@Table( name = "tweets")
public class Tweet {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @NotBlank
    @Size(max = 140)
    private String tweet;

    public Tweet() {}

    public Tweet (String tweet) {
        this.tweet = tweet;
    }

    // getters and setters
    public Long getId() {
        return id;
    }

    public void setId(Long id) {
        this.id = id;
    }

    public String getTweet() {
        return tweet;
    }

    public void setTweet(String tweet) {
        this.tweet = tweet;
    }
}
```

Create Tweet Controller:

\$ vi controllers/TweetController.java

```
package com.postgres.demopg.controllers;
import java.util.Optional;
import jakarta.validation.Valid;
import org.apache.tomcat.util.net.openssl.ciphers.Authentication;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.postgres.demopg.models.Tweet;
import com.postgres.demopg.repository.TweetRepository;
import org.springframework.data.domain.Page;
import org.springframework.data.domain.Pageable;

@CrossOrigin(origins = "*", maxAge = 3600)
@RestController
@RequestMapping("/api/tweets")

public class TweetController {
    @Autowired
    private TweetRepository tweetRepository;

    @GetMapping("")
    public Page<Tweet> getTweet(Pageable pageable) {
        return tweetRepository.findAll(pageable);
    }

    @PostMapping("")
    public Tweet createTweet(@Valid @RequestBody Tweet tweet) {
        Tweet myTweet = new Tweet(tweet.getTweet());
        tweetRepository.save(myTweet);
        return myTweet;
    }
}
```

Create Tweet Repository:

\$ vi repository/TweetRepository.java

```
package com.postgres.demopg.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.postgres.demopg.models.Tweet;

@Repository
public interface TweetRepository extends JpaRepository<Tweet, Long> {

}
```

\$ mvn spring-boot:run

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ sudo -u postgres psql
psql (17.9 (Ubuntu 17.9-0ubuntu0.25.10.1))
```

```
Type "help" for help.
```

```
postgres=# \list
```

```
postgres=# \connect user_api
```

```
You are now connected to database "user_api" as user "postgres".
```

```
user_api=# \dt
```

List of relations

Schema	Name	Type	Owner
-----+-----+-----+-----			
public	tweets	table	postgres

```
(3 rows)
```

```
user_api=# select * from tweets;
```

```
id | tweet
```

```
----+-----
```

```
(0 rows)
```

```
user_api=#
```

get method



http://localhost:8080/api/tweets



Save



</>

GET



http://localhost:8080/api/tweets

Send



Params

Authorization

Headers (6)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies

Headers (8)

Test Results

200 OK

12 ms

734 B



Pretty

Raw

Preview

Visualize

JSON



```
3  {
4    "tweet": "adsoft",
5    "id": 1
6  },
7  {
8    "tweet": "adsoftsito is my idol",
9    "id": 2
10 },
11 {
12   "tweet": "i hate windows",
13   "id": 3
14 },
15 {
16   "tweet": "i hate office",
17   "id": 4
18 },
19 {
20   "tweet": "i love linux",
21   "id": 5
22 }
```

Runner

Capture requests

Cookies

Trash



post method



http://localhost:8080/api/tweets

Save

</>

POST

http://localhost:8080/api/tweets

Send

Params Authorization Headers (8) **Body** Scripts Settings

Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON**

Beautify

```
1 {  
2   "tweet": "i love linux"  
3 }  
4  
5
```

Body Cookies Headers (8) Test Results

200 OK • 21 ms • 284 B • 🌐 | ⋮

Pretty

Raw

Preview

Visualize

JSON



```
1 {  
2   "tweet": "i love linux",  
3   "id": 5  
4 }
```

Flutter client - native

\$ flutter create tweeter

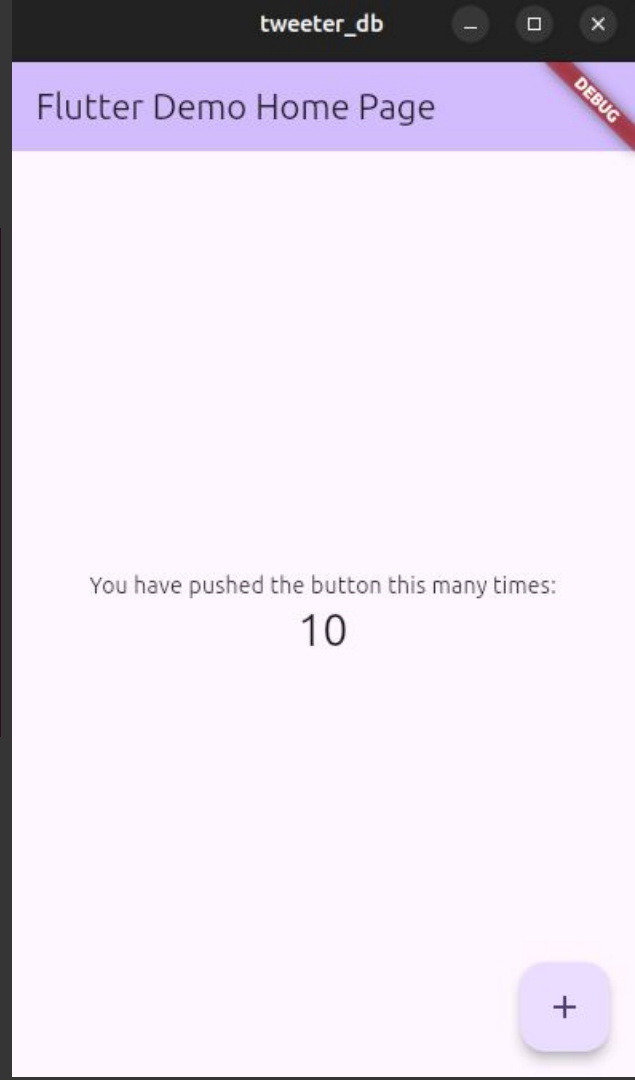
\$ cd tweeter

\$ vi pubspec.yaml

```
dependencies:  
  flutter:  
    sdk: flutter  
  
  # The following adds the Cupertino Icons font to your application.  
  # Use with the CupertinoIcons class for iOS style icons.  
  cupertino_icons: ^1.0.8  
  http: ^1.1.0  
  
dev_dependencies:  
  flutter_test:  
    sdk: flutter
```

\$ flutter pub get

\$ flutter run



Flutter client - chrome

\$ flutter run -d chrome

Flutter client - android

\$ flutter devices

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/database/tweeter_db$ flutter devices
Found 2 connected devices:
  Linux (desktop) • linux • linux-x64 • Ubuntu 25.10 6.17.0-22-generic
  Chrome (web) • chrome • web-javascript • Google Chrome 147.0.7727.101
```

Device emulator-5554 is not authorized.

You might need to check your device for an authorization dialog.

Run "flutter emulators" to list and start any available device emulators.

If you expected another device to be detected, please run "flutter doctor" to diagnose potential issues.

You may also try increasing the time to wait for connected devices with the "--device-timeout" flag.

Visit <https://flutter.dev/setup/> for troubleshooting tips.

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/database/tweeter_db$
```

\$ flutter build apk

\$ flutter install

Flutter client

```
$ cd lib
```

```
$ mkdir models
```

```
$ mkdir services
```

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/lib$ ls -la
total 24
drwxrwxr-x  4 adsoft adsoft 4096 Mar  8 13:44 .
drwxrwxr-x 13 adsoft adsoft 4096 Mar  8 13:43 ..
-rw-rw-r--  1 adsoft adsoft 4132 Mar  8 14:15 main.dart
drwxrwxr-x  2 adsoft adsoft 4096 Mar  8 13:53 models
drwxrwxr-x  2 adsoft adsoft 4096 Mar  8 13:44 services
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/lib$
```

Flutter client

\$ vi models/tweet.dart

```
class Tweet {
  final int id;
  final String tweet;

  Tweet({
    required this.id,
    required this.tweet,
  });

  /// Convert JSON to Tweet object
  factory Tweet.fromJson(Map<String, dynamic> json) {
    return Tweet(
      id: json['id'] ?? '',
      tweet: json['tweet'] ?? '',
    );
  }

  /// Convert Tweet object to JSON
  Map<String, dynamic> toJson() {
    return {
      'id': id,
      'tweet': tweet,
    };
  }

  @override
  String toString() => 'Tweet(id: $id, tweet: $tweet)';
}
```

Flutter client

\$ vi models/tweet_response.dart

```
import 'tweet.dart';

/// Model for the paginated response from the tweets API endpoint
class TweetResponse {
  final List<Tweet> content;

  TweetResponse({
    required this.content,
  });

  /// Convert JSON to TweetResponse object
  factory TweetResponse.fromJson(Map<String, dynamic> json) {
    final contentList = json['content'] as List<dynamic>? ?? [];
    return TweetResponse(
      content: contentList
        .map((tweet) => Tweet.fromJson(tweet as Map<String, dynamic>))
        .toList(),
    );
  }

  @override
  String toString() =>
    'TweetResponse(content: ${content.length}, totalElements)';
}
```

Flutter client

\$ vi services/tweet_service.dart

```
import 'package:http/http.dart' as http;
import 'dart:convert';
import '../models/tweet.dart';
import '../models/tweet_response.dart';

/// Singleton service for managing REST API calls to the tweets endpoint
/// Implements the Singleton pattern to ensure only one instance exists
class TweetService {
  static final TweetService _instance = TweetService._internal();

  final String baseUrl = 'http://localhost:8080/api';
  late http.Client _httpClient;

  // Private constructor
  TweetService._internal() {
    _httpClient = http.Client();
  }

  /// Factory constructor that always returns the same instance
  factory TweetService() {
    return _instance;
  }

  /// Get the singleton instance
  static TweetService getInstance() {
    return _instance;
  }

  /// Fetch all tweets from the API
  Future<List<Tweet>> fetchTweets() async {
    try {
      final response = await _httpClient.get(
        Uri.parse('$baseUrl/tweets'),
      );

      if (response.statusCode == 200) {
        final jsonData = jsonDecode(response.body) as Map<String, dynamic>;
        final tweetResponse = TweetResponse.fromJson(jsonData);
        return tweetResponse.content;
      } else {
        throw Exception(
          'Failed to load tweets. Status code: ${response.statusCode}',
        );
      }
    } catch (e) {
      throw Exception('Error fetching tweets: $e');
    }
  }

  /// Close the HTTP client (cleanup)
  void dispose() {
    _httpClient.close();
  }
}
```

Flutter client

\$ vi lib/main.dart

<https://github.com/adsoftsito/principios-design/blob/main/tweeter/main.dart>

Flutter client - chrome



Deploy REST API - postgresql cloud



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 Continue with email

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Deploy REST API - postgresql cloud

Project: **adsoft** Organization: **My Organization**

 You can select a specific cloud provider and region on the Professional tier.

Asia Pacific

Australia

Europe

North America

Plan

	Plan	VMs	CPUs per VM	RAM per VM	Storage	Monthly price
☒	Free-1-1gb	1	1	1 GB	1 GB	Free

Service basics

Name*

The service name cannot be changed afterwards

pg-adsoftsitod

Version

You can select the version on the Professional tier

PostgreSQL 17

[⊕ Add tag to this service](#)

Service summary

Service

PostgreSQL 17

Name _____

pg-adsoftsito

Service tier

Free

Cloud

North America

Plan

Free-1-1qb

Monthly price

Free

Create service

Deploy REST API - postgresql cloud

The screenshot displays the Aiven console interface for a PostgreSQL service. The top navigation bar includes the Aiven logo, a breadcrumb trail, and links for Home, Projects, Tools, Billing, and Admin. The left sidebar contains a list of navigation options, with 'Overview' selected. The main content area is titled 'Connection information' and features a 'Quick connect' button. Below this, a table lists various connection parameters.

Connection information		Quick connect
Service URI	postgres://CLICK_TO_REVEAL_PASSWORD@pg-adsoftsito-adsoft.g.aivencloud.com:13078/defaultdb?sslmode=require	
Database name	defaultdb	
Host	pg-adsoftsito-adsoft.g.aivencloud.com	
Port	13078	
User	avnadmin	
Password	*****	
SSL mode	require	
CA certificate	Show	
Connection limit	20	

Deploy REST API - build

```
$ rm target/*.jar
```

```
$ mvn clean package
```

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ ls target/*.jar  
target/demopg-0.0.1-SNAPSHOT.jar  
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$
```

Deploy REST API - Dockerfile

\$ vi Dockerfile

```
FROM openjdk:22-rc-oracle
```

```
LABEL maintainer="adsoft@live.com.mx"
```

```
VOLUME /tmp
```

```
EXPOSE 8080
```

```
ARG JAR_FILE=target/SpringBootSecurityPostgresqlApplication-0.0.1-SNAPSHOT.jar
```

```
ADD ${JAR_FILE} antlr-api.jar
```

```
#RUN
```

```
ENTRYPOINT ["java", "-Djava.security.egd=file:/dev/./urandom", "-jar", "/antlr-api.jar"]
```

Deploy REST API - docker build

```
$ sudo docker build -t <your-docker-hub-user>/tweeter-api .
```

NOTE: don't forget the point

```
adsoft@fnt-ori213: ~/Desktop/2026/usbi/principios/patterns/creational/singleton/demogg$ sudo docker build -t adsoft/tweeter-api .
[sudo: authenticate] Password:
[+] Building 298.3s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 304B                                0.0s
=> [internal] load metadata for docker.io/library/openjdk:22-rc-oracle 2.6s
=> [auth] library/openjdk:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                  0.4s
=> => transferring context: 54.55MB                                   0.4s
=> [1/2] FROM docker.io/library/openjdk:22-rc-oracle@sha256:08b2d714025cbba08c787f5395d931ba 295.0s
=> => resolve docker.io/library/openjdk:22-rc-oracle@sha256:08b2d714025cbba08c787f5395d931bae8 0.0s
=> => sha256:972029ff9873af36c6c0fcee05b14acbc57a61ecd0b8bf86b167bdf46f973 48.98MB / 48.98MB 140.6s
=> => sha256:d04e36430406bb6588fe835b6268588b825205320c94fbd96651f340a618d 37.73MB / 37.73MB 112.6s
=> => sha256:07c9b018c642931f767eff0ee01c56b4805e644ebaff9ba39c4e2ce5d1f 202.77MB / 202.77MB 292.8s
=> => sha256:08b2d714025cbba08c787f5395d931bae89345a856e4ab1be20891b8926eed82 2.58kB / 2.58kB 0.0s
=> => sha256:aa232ec30b86617a044994b3898a1a61d7ee739cb6aae7944084c37b0aa1255b 1.57kB / 1.57kB 0.0s
=> => sha256:2a271298cf196c7aaa694279395e0c7e10640b40f3071042ffd3a0740c1598f1 3.58kB / 3.58kB 0.0s
=> => extracting sha256:972029ff9873af36c6c0fcee05b14acbc57a61ecd0b8bf86b167bdf46f973823 1.3s
=> => extracting sha256:d04e36430406bb6588fe835b6268588b825205320c94fbd96651f340a618da24 1.1s
=> => extracting sha256:07c9b018c642931f767eff0ee01c56b4805e644ebaff9ba39c4e2ce5d1f174e7 2.1s
=> [2/2] ADD target/demogg-0.0.1-SNAPSHOT.jar antlr-api.jar      0.3s
=> exporting to image                                              0.2s
=> => exporting layers                                              0.2s
=> => writing image sha256:824bc5ea1f184f5e20095d570a687e00b3b3a6be16d1c52c4e30ac8c1e7cf35b 0.0s
=> => naming to docker.io/adsoft/tweeter-api                      0.0s
```

Deploy REST API - docker build

```
$ sudo docker images
```

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ sudo docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
adsoft/tweeter-api	latest	824bc5ea1f18	About a minute ago	612MB
adsoft/flutter	latest	928a75849ce1	13 days ago	147MB
adsoft/site	latest	e36eb036737a	2 weeks ago	140MB
httpd	latest	d2825031e94c	5 weeks ago	117MB
hello-world	latest	1b44b5a3e06a	7 months ago	10.1kB

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$
```

Deploy REST API - docker build

```
$ sudo docker run --name test-api -p 8080:8080 -d adsoft/tweeter-api
```

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ sudo docker run --name test-api -p 8080:8080 -d adsoft/tweeter-api
```

```
25dfb7a03325b63a48b6fd3859d70a6cae2c77e2f9870b32cf35198fdb808913
```

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAME
25dfb7a03325	adsoft/tweeter-api	"java -Djava.securit..."	5 seconds ago	Up 5 seconds	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp	test-api

Deploy REST API - test

localhost:8080/api/tweets

Pretty-print ☒

```
{
  "content": [
    {
      "tweet": "i love linux - from cloud",
      "id": 1
    }
  ],
  "empty": false,
  "first": true,
  "last": true,
  "number": 0,
  "numberOfElements": 1,
  "pageable": {
    "offset": 0,
    "pageNumber": 0,
    "pageSize": 20,
    "paged": true,
    "sort": {
      "empty": true
    }
  }
}
```

Deploy REST API - deploy docker hub

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$ sudo docker push adsoft/tweeter-api
Using default tag: latest
The push refers to repository [docker.io/adsoft/tweeter-api]
ebb8cec762e7: Pushed
6acaaba9e97a: Mounted from library/openjdk
cf3ce83da20a: Mounted from library/openjdk
0a628c3f1dfa: Mounted from library/openjdk
latest: digest: sha256:ca77aed29ce4e7f2f64f47c7bb547a30d3b4d87bf85c14cfbfece77c6678a8d8 size: 1166
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/demopg$
```

The screenshot shows the Docker Hub web interface. The top navigation bar is blue with the Docker Hub logo, 'Explore' and 'My Hub' tabs, a search bar, and a 'CtrlK' button. The left sidebar shows the user profile 'adsoft' (Docker Personal) and a list of links: 'Repositories' (selected), 'Hardened Images', 'Collaborations', 'Settings', 'Default privacy', and 'Notifications'. The main content area is titled 'Repositories' and states 'All repositories within the adsoft namespace.' Below this is a search bar and a dropdown menu set to 'All content'. A 'Create a repository' button is in the top right. A table lists the repositories:

Name	Last Pushed	Contains	Visibility	Sco
adsoft/tweeter-api	2 minutes ago	IMAGE	Public	Inac

Deploy REST API - deploy render

A adsoftsito's Works...

↕

🌐 New Web Service

▼

🔍 Search

^ K

+ New

⚡ Upgrade

?

A

New Web Service

Source Code

Git Provider

Public Git Repository

Existing Image

Image URL

Deploy an image from a Docker registry

🐳

adsoft/tweeter-api

✓

Credential (Optional)

No credential

▼

Connect →

In flutter - update manifest

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/android/app/src/main$ pwd
/home/adsoft/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/android/app/src/main
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/android/app/src/main$ ls -la
total 24
drwxrwxr-x  5 adsoft adsoft 4096 Mar  8 13:38 .
drwxrwxr-x  5 adsoft adsoft 4096 Mar  8 13:38 ..
-rw-rw-r--  1 adsoft adsoft 2195 Mar  8 13:38 AndroidManifest.xml
drwxrwxr-x  3 adsoft adsoft 4096 Mar  8 13:38 java
drwxrwxr-x  3 adsoft adsoft 4096 Mar  8 13:38 kotlin
drwxrwxr-x 11 adsoft adsoft 4096 Mar  8 13:38 res
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/android/app/src/main$
```

In flutter - update manifest

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
```

```
<uses-permission android:name="android.permission.INTERNET" />
```

```
<application
```

```
    android:label="tweeter"
```

```
    android:name="${applicationName}"
```

```
    android:icon="@mipmap/ic_launcher">
```

```
    <activity
```

In flutter - update services with cloud REST API

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/lib/services$ pwd
/home/adsoft/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/lib/services
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/lib/services$ ls -la
total 12
drwxrwxr-x 2 adsoft adsoft 4096 Mar 10 10:28 .
drwxrwxr-x 4 adsoft adsoft 4096 Mar  8 13:44 ..
-rw-rw-r-- 1 adsoft adsoft 1466 Mar 10 09:46 tweet_service.dart
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter/lib/services$ vi tweet_service.d
art
```

In flutter - update services with cloud REST API

```
import '../models/tweet.dart';
import '../models/tweet_response.dart';

/// Singleton service for managing REST API calls to the tweets endpoint
/// Implements the Singleton pattern to ensure only one instance exists
class TweetService {
  static final TweetService _instance = TweetService._internal();

  final String baseUrl = 'https://adsoftsito-api.render.com/api';
  // final String baseUrl = 'http://localhost:8080/api';
  late http.Client _httpClient;
```

In flutter - build apk

```
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter$ pwd
/home/adsoft/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter$ ls
README.md          android ios  linux  pubspec.lock  test      web
analysis_options.yaml  build   lib  macos  pubspec.yaml  tweeter.iml  windows
adsoft@fnt-ori213:~/Desktop/2026/usbi/principios/patterns/creational/singleton/tweeter$ flutter build apk
Resolving dependencies...
Downloading packages...
> characters 1.4.1 (was 1.4.0)
> matcher 0.12.19 (was 0.12.17)
> material_color_utilities 0.13.0 (was 0.11.1)
  meta 1.17.0 (1.18.2 available)
> test_api 0.7.10 (was 0.7.7) (0.7.11 available)
  vector_math 2.2.0 (2.3.0 available)
Changed 4 dependencies!
3 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
Running Gradle task 'assembleRelease'...
```



In spring-boot

add CORS, delete

```
import com.postgres.demopg.models.Tweet;
import com.postgres.demopg.repository.TweetRepository;

import org.springframework.data.domain.Page;
import org.springframework.data.domain.Pageable;

import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.CrossOrigin;

@CrossOrigin(origins = "*", maxAge = 3600)
@RestController
@RequestMapping("/api/tweets")

public class TweetController {

    @Autowired
    private TweetRepository tweetRepository;

    @GetMapping("")
    public Page<Tweet> getTweet(Pageable pageable) {
        return tweetRepository.findAll(pageable);
    }

    @PostMapping("")
    public Tweet createTweet(@Valid @RequestBody Tweet tweet) {

        Tweet myTweet = new Tweet(tweet.getTweet());
        tweetRepository.save(myTweet);
        return myTweet;
    }

    @DeleteMapping("/{id}")
    public void deleteTweet(@PathVariable Long id) {
        tweetRepository.deleteById(id);
    }
}
```

Test delete in postman

Overview

GET https://swapi.dev/ap

GET http://localhost:8080

POST http://localhost:330

DEL http://localhost:330

DEL http://localhost:8080

+

▼

No environment

▼



http://localhost:8080/api/tweets/6

Save

▼

DELETE

▼

http://localhost:8080/api/tweets/6

Send

▼

Docs

Params

Authorization

Headers (6)

Body

Scripts

Settings

Cookies

☒ none

☐ form-data

☐ x-www-form-urlencoded

☐ raw

☐ binary

☐ GraphQL

This request does not have a body

Body

Cookies

Headers (7)

Test Results



200 OK

94 ms

212 B



...

Raw

▼

Preview

Visualize

▼



1

In flutter - update from:

